
CLOSED-DOMAIN RAG • CHAT-HISTORY-ONLY KNOWLEDGE

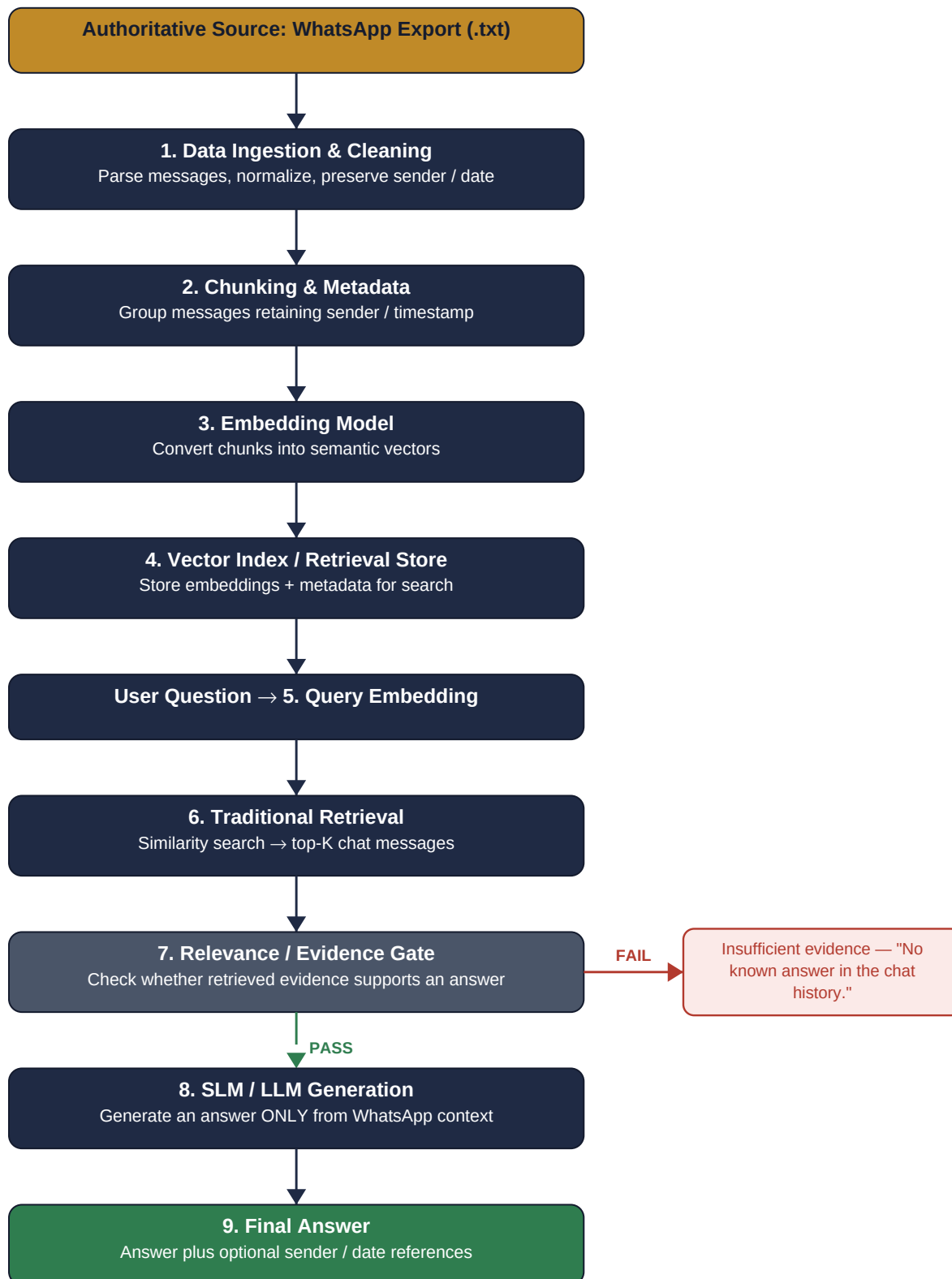
WhatsApp Group Chat Q&A Chatbot

Embedding + Traditional Retrieval + SLM / LLM

1. Objective

Build a question-answering chatbot whose knowledge is **strictly limited to a WhatsApp group chat history**. The chatbot should answer only when relevant evidence exists in the chat history; otherwise it must return “No known answer in the chat history.”

2. Final Architecture Pipeline



3. Component Responsibilities & Knowledge Boundary

Component	Purpose
WhatsApp History	The single source of truth / knowledge base.
Data Processing	Clean and structure the raw WhatsApp export without changing factual content.
Embedding Model & Vector Index	Enable semantic matching and efficiently retrieve historical messages.
Relevance Gate	Prevent unsupported questions from reaching the generation stage.
SLM / LLM	Understand retrieved context and formulate a natural-language answer.

*The SLM/LLM is **not** the knowledge source. Decision rule: if the retrieval/evidence layer cannot establish that the answer is present in the chat history, the system must not generate an answer from general model knowledge.*

4. Example Flow

Stage	Content
User Question	When is the deployment scheduled?
Retrieved History	Rahul: The deployment is scheduled for Friday at 6 PM.
Evidence Gate	Sufficient evidence found.
Final Answer	The deployment is scheduled for Friday at 6 PM.